MILITARY SPECIFICATION

FUZE, ROCKET, PIBD, M438
SUB-ASSEMBLY WITH DETONATOR

This Amendment forms a part of Military Specification MIL-F-48057B (AR), dated 17 February 1981, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 5

3.4.2.2: Delete "ambient temperature" and substitute "any temperature between -40°F and +160°F".

3.5.4: Delete in its entirety and substitute:

"3.5.4 Capacitor discharge test (Type II only). The capacitor, when charged to 22.5 ± 0.5 volts, shall provide a detonator fire pulse having an amplitude of 15 volts minimum when monitored after 30 seconds (minimum). The monitoring is accomplished by gating the SCR with a voltage of 22.5 ± 0.5 volts. The 15 volt minimum requirement shall apply at any temperature between -40°F and +160°F."

PAGE 6

3.8.8: Delete in its entirety.

PAGE 8

4.3.1:

Delete "11830039" and substitute "9378352".
Delete "11706267" and substitute "8883713" in two places.
Delete "11008392" and substitute "8883718" in two places.
Delete "11001692" and substitute "8883710" in two places.
Delete "11008394" and substitute "9260774".
Delete "11001687" and substitute "8883703" in two places.
Delete "11008395" and substitute "9260804" in two places.
Delete "11001689" and substitute "8883700" in two places.

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Delete item "Outer Plate Assembly" in its entirety.
Delete "11706262" and substitute "9260776" in two places.
Delete "10" from Quantity for Escapement Assembly and Inner Plate Assembly and substitute "20".
Delete "80" from Quantity for Safety & Arming Device and substitute "120".

PAGE 9

* 4.3.1: Delete "25" from Quantity for Magnetic Impulse Generator Assembly, and substitute "30". Delete "32" from Sub-Assy w/Detonator and Fuze Components, and substitute "54".

Add: "Stud, Spring 8883721 8883721 10".

* Item List, Add under following headings:

Part Description	Type I	Type II	Quantity
"Safety & Arming Device, Less Power Supply Striker Assembly	11706290		25
Guide, Bushing Assembly		9386917	15"

PAGE 10

Table I: Add to Note b "(15) and +160°F (15)".

PAGE 11

TABLE I: Delete "S&A Position Check" in its entirety. In No. of Sample Units, delete "16" and substitute "27" in two places.

PAGE 12

TABLE I: Delete "Dwg. 11830039" and substitute "Dwg. 9378352".

PAGE 14

TABLE I:

Delete "Dwg. 11001692" and substitute "Dwg. 8883710".
Delete "Dwg. 11001687" and substitute "Dwg. 8883703".
Delete "Dwg. 11001689" and substitute "Dwg. 8883700".
Delete "Dwg. 11008394" and substitute "Dwg. 9260774".
Delete "Push Off Test" and substitute "Load Test", 3 Places.
Delete "10" and substitute "20" for Escapement Assembly, 2 places.

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PAGE 15

TABLE 1: Delete "Dwg. 11008395" and substitute "Dwg. 9260804". Delete "10" and substitute "20", 2 places.

PAGE 16

TABLE 1: Delete "Examination and Test" for Outer Plate Assembly (Dwg. 11008400) in its entirety.

Delete "Dwg. 11706262" and substitute "Dwg. 9260776".
Delete "Shaft Torque Test" and "Pallet Lever Push Off Test" in their entirety and substitute the following three tests:

"Load Test of Lever Shaft	10	3.2	4.5.5.13"
"Load Test of Lever Pallet	10	3.2	4.5.5.8"
"Torque Test of Lever Shaft	10	3.2	4.5.5.7"

Delete "Non-Arm Test" in its entirety.

In No. of Sample Units, Delete "80" and substitute "120" in four places; Delete "40" and substitute "60(a,b)" in two places.

In Arm Time (27 g's) Test, Delete "4.5.11.3" and substitute "4.5.11.4".

In Arm Time (40 g's) Test, Delete "4.5.11.4" and substitute "4.5.11.5".

In Non-Arming Test, Delete "4.5.11.3" and substitute "4.5.11.2.1 or 4.5.11.2.2".

Add: Notes "a. 20 at 70°F ± 5°F; 20 at -40°F ± 5°F, 20 at 160°F ± 5°F" and "b: Temperature conditioning shall be 4 hours minimum, and each S&A shall be tested within 3 minutes of removal from extreme temperatures. Be advised that removing the S&A from the cold temperature environment prior to performing the centrifuge test may cause frost to form on the metal parts. This may have an effect on the test results due to attenuation of the acoustic signal of the arming mechanism, thus making it difficult to distinguish the arm signal from background noise. The contractor shall take appropriate action to alleviate this potential problem. One effective method has been to seal each S&A in a plastic bag prior to temperature conditioning. Bagged S&A's are then

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subjected to the centrifuge test. Another method has been to house the centrifuge inside the temperature conditioning chamber. S&A's are then tested inside the temperature conditioning chamber."

PAGE 17

* TABLE I, Add:

"Safety & Arming Device, Less Power Supply Striker Assembly (Dwg. 11706290) Examination for defects	25	3.2	4.4.2.26
Guide, Bushing Assembly (Dwg. 9386917) Examination for defects	15	3.2	4.4.2.32".

PAGE 18

TABLE 1: In No. of Sample Units, Delete "25" and substitute "30" in three places.

Delete "Dwg. 11706267" and substitute "Dwg. 8883713".
Delete "Dwg. 11008392" and substitute "Dwg. 8883718".
Add "Stud Spring, (Dwg. 8883721) Examination for Defects 10 3.2 4.4.2.24".

PAGE 25

4.4.2.6: Delete "Critical 3" in its entirety. Add "(Type I Only)" to Major 105 Examination.

PAGE 26

4.4.2.7: In No. of Sample Units, Delete "16" and substitute "32" in two places.

PAGE 27

4.4.2.8:

Delete "11706267" and substitute "8883713" in two places.
Delete "11008394" and substitute "9260774".

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PAGE 34

4.4.2.15: Delete "11830039" and substitute "9378352".

PAGE 36

4.4.2.17: Delete "11001692" and substitute "8883710" in two places.
Delete "11008394" and substitute "9260774".
Delete Critical 1 in its entirety and substitute Critical 1 and Critical 2: "1 Maximum gap between gear and pinion 100% 3.2 Gage" and "2 Load test of excapement gear and pinion assembly 100% 3.2 4.5.5.6".

PAGE 37

4.4.2.18: Delete "11001687" and substitute "8883703" in two places.
Delete "11008394" and substitute "9260774".

Delete Critical 1 in its entirety and substitute Criticals 1 and 2:

"1	Maximum gap between gear and pinion	100%	3.2	Gage"
"2	Load test of No. 1 gear and pinion Assembly	100%	3.2	4.5.5.4"

PAGE 38

4.4.2.19: Delete "11001689" and substitute "8883700" in two places.
Delete "11008394" and substitute "9260774".
Delete Critical 1 in its entirety and substitute Criticals 1 and 2:

"1	Maximum gap between gear and pinion	100%	3.2	Gage"
"2	Load test of No. 2 gear and pinion Assembly	100%	3.2	4.5.5.5"

PAGE 39

4.4.2.20: Delete "11706262" and substitute "9260776" in two places.
Delete "11008394" and substitute "9260774".

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Delete Critical 1 in its entirety and substitute Criticals 1, 2 and 3:

"1	Maximum gap between lever and shaft	100%	3.2	Gage"
"2	Load test of lever shaft	100%	3.2	4.5.5.13"
"3	Load test of lever pallet	100%	3.2	4.5.5.8"

Delete "None Defined" and add Major 101:

"101 Torque test of lever shaft 100% 3.2 4.5.5.7".

PAGE 41

4.4.2.22: Delete "11008394" and substitute "9260774".

PAGE 42

4.4.2.23: Delete "11008395" and substitute "9260804" in two places.
Delete "11008394" and substitute "9260774".

PAGE 43

4.4.2.24: Delete in its entirety and substitute included Page 7 (Page 43).

PAGE 44

4.4.2.25: Delete Majors "107" and "109" in their entirety.

Major 110, Delete "4.5.11.4" and substitute "4.5.11.5".

Major 111, Delete "4.5.11.3" and substitute "4.5.11.4".

* Next Higher Assembly Block, Delete "11830058" and substitute "11830062".

PAGE 46

* 4.4.2.26, Title, Delete "(Type I) or Release Pin (Type II)" and substitute "(Type I only)".

Delete "Type I Type II".

Drawing Number Block, Delete "11830058".

Next Higher Assembly Block, Delete "11830062".

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QUALITY CONFORMANCE INSPECTION

MIL-F-48057B (AR)
AMENDMENT 7

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Stud, Spring				8883721
					NEXT HIGHER ASSEMBLY 11008389
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR 101	Diameter if stud, min.		0.40%	3.2	Gage
MINOR 201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

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4.4.2.27: Major 101, Delete "(a)"; delete "4.5.11.3" and substitute "4.5.11.2.1 or 4.5.11.2.2".

Add "Major 103 Release pin improperly assembled (Type II Only) 100% 3.2 Visual".

PAGE 47

* 4.4.2.27, Delete in its entirety and substitute included Page 9 (Page 47)

PAGE 48

4.4.2.29: Delete Major 101 in its entirety. Major 102: Delete "14.5.12.2" and substitute "4.5.12.2".

PAGE 49

4.4.2.30: In No. of Sample Units, Delete "15" and substitute "30".

PAGE 50

4.4.2.31: Delete "11008392" and substitute "8883718" in two places.

* Add new paragraph "4.4.2.32" included page 10 (page 50A).

PAGE 51

4.4.4.1.1: Delete "S&A position check" and substitute "Non-Arming Test".

4.4.4.1.2: Delete "4.5.10".

PAGE 52

4.4.4.2: Under VERIFIED INSPECTION, Delete "4.5.10 - S&A Position Check" and substitute "4.5.11.2 - Non-Arming Test".

PAGE 53

4.5.4: Change paragraph "c" to paragraph "d" and add new paragraph "c" as follows:

"c. A Power Supply and a simulated detonator load (2 to 5 ohms) are required to perform the capacitor discharge test."

4.5.5.1: Delete "Apply a" and substitute "Apply the required". Delete "resistive" and substitute "required".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER Type I / Type II
4.4.2.27	Safety & Arming Device Assembly				11008351 / 11830062
					NEXT HIGHER ASSEMBLY
					9259237 / 11830059
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Non-arming		100%	3.3.1	4.5.11.2.1/4.5.11.2.2
102	Power supply striker improperly assembled (Type I only)		100%	3.2	Gage
103	Release pin improperly assembled (Type II only)		100%	3.2	Visual
104	Detonator improperly assembled (Type II only)		0.40%	3.2	Gage & Visual
105	Circuit continuity (Type II only)		100%	3.3.4	4.5.8
106	Detonator clip protrudes beyond outer periphery of rotor after assembly (Type II only)		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.32	Guide, Bushing Assembly (Type II only)		9386917
			NEXT HIGHER ASSEMBLY 11830055

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR 101	Staking improper		0.40%	3.2	Visual
MINOR 201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRSM [illegible] (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.5.5.2: Delete "Apply a" and substitute "Apply the required". Delete "resistive" and substitute "required".

4.5.5.3: Delete "Apply a" and substitute "Apply the required". Delete "resistive" and substitute "required".

PAGE 54

4.5.5.4: Delete in its entirety and substitute:

"4.5.5.4 Load test of No. 1 gear and pinion assembly. Apply the required axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The required force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 1 gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot."

4.5.5.5: Delete in its entirety and substitute:

"4.5.5.5 Load test of No. 2 gear and pinion assembly. Apply the required axial load perpendicular to the No. 2 gear in such a direction as to test the assembly of the No. 2 gear and pinion. The required force shall act against the No. 2 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 2 gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot."

4.5.5.6: Delete in its entirety and substitute:

"4.5.5.6 Load test of escapement gear and pinion assembly. Apply the required axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The required force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any escapement gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot."

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4.5.5.7: Delete in its entirety and substitute:

"4.5.5.7 Torque test of lever shaft. A torque shall be applied between the lever and the lever shaft. Any lever shaft that does not comply with the drawing requirement shall be classed defective and removed from the lot."

4.5.5.8: Delete in its entirety and substitute:

"4.5.5.8 Load test of lever pallet. Apply the required axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The required force shall act against the lever. Upon removal of the required force, the height of the lever pallet above the lever shall be inspected for compliance with the drawing requirement. Any lever pallet that fails the requirement shall be classed defective and removed from the lot."

4.5.5.9: Delete in its entirety and substitute:

"4.5.5.9 Load test of pillars. Apply the required axial load perpendicular to the inner plate and in such a direction as to test the assembly of the pillars to the inner plate and outer plate. The required force shall act against the inner plate. Any pillars that do not comply with the drawing requirement shall be classed defective."

4.5.5.10: Delete in its entirety and substitute:

"4.5.5.10 Load test of bridge rivet. Apply the required load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The required force shall act against the inner plate. Any bridge rivet that does not comply with the drawing requirement shall be classed defective."

4.5.5.11: Delete "Apply a" and substitute "Apply the required". Delete "resistive" and substitute "required".

4.5.5.12: Delete "Apply a" and substitute "Apply the required". Delete "resistive" and substitute "required".

Add 4.5.5.13:

"4.5.5.13 Load test of lever shaft. Apply the required axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and shaft. The required

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force shall act against the lever. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any lever shaft that fails the requirement shall be classed defective and removed from the lot."

PAGE 55

4.5.10: Delete in its entirety.

PAGE 56

4.5.10: Delete paragraph starting with "Defect standards ..." from para 4.5.10 and add it to para 4.5.11.2.1.

4.5.11.1: Delete last four lines and substitute "specified "g" limits for the arming tests are reached. This test equipment, if housed within a temperature conditioning chamber, shall be capable of operation between -40°F and +160°F with a tolerance of ± 5°F. The acceleration tolerance shall be ± .10g. Time measuring equipment shall be accurate within ± 0.01 second."

4.5.11.2.1: Delete "containing a power supply striker and spring but no explosives" and substitute "completely assembled".

4.5.11.2.2: Delete in its entirety, except for title, and substitute "The device completely assembled shall be placed in the centrifuge and tested at 12g for five seconds. The device shall be tested in a spring-loaded fixture which simulates the MIG spring. The device shall meet the requirements of 3.3.1. Any S&A that fails to meet the requirement shall be classed defective."

4.5.11.3: Delete in its entirety.

4.5.11.4: Delete "4.5.11.2.1 or 4.5.11.2.2 except that the" and substitute "4.5.11.1. The".

PAGE 57

4.5.11.5: Delete "4.5.11.2.1 or 4.5.11.2.2 except that the" and substitute "4.5.11.1. The".

Add "and removed from the lot" to last sentence.

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4.5.12.1: Delete second line and substitute "required in 3.4.2.3. Following helicopter vibration, the assemblies shall be temperature conditioned for a minimum of four hours, 15 at -40°F ± 5°F and 15 at +160°F ± 5°F. They shall then be assem-".

Add "One defective shall be cause for lot rejection."

PAGE 58

4.5.17: Delete in its entirety, except for title, and substitute "Following the helicopter vibration of 4.5.15, the units shall be temperature conditioned for a minimum of four hours, 24 at -40°F ± 5°F and 24 at +160°F ± 5°F. They shall then be tested as in 4.5.11.1. Arming time shall start upon setback weight release and shall stop at rotor detent. The arming time shall be recorded to the nearest millisecond using an acoustic pick-up which feeds the arming cycle noise signal to the recorder."

TABLE II: For "Over Max", delete "0" and "1" and substitute "1" and "2", respectively.

For "Firing", delete "0" and "1" and substitute "1" and "2", respectively.

Add "Note: If one reject occurs, the lot shall be accepted. If three or more rejects occur, the lot shall be rejected. If one reject occurs in both arming time over max and firing, a second sample of 48 subassemblies shall be tested. If the combined number of rejects in both samples is three or more, the lot shall be rejected."

PAGE 59

4.5.18: Delete in its entirety and substitute the following:

"4.5.18 Fuze function test. A random sample of 64 fuzes shall be selected from each fuze assembly lot for this test. The sub-assemblies shall be sent to a Government designated facility to be loaded with a booster assembly in accordance with the latest drawings and specifications. The fuzes shall then be assembled to the appropriate rounds and fired from a 2.75 Inch Rocket Launcher or equivalent situated on the ground to impact on natural terrain at a distance of 3000 meters. If three defects are found, the lot shall be accepted. If four or more fuzes fail to comply with the requirement specified, the lot shall be rejected."

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6.3: Delete "US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I" and substitute "Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAF-I(D)".

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